# Solution Requirements

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| **Date** | 22 June 2026 |
| **Team ID** | |
| **Project Name** | EduGenie |
| **Maximum Marks** | 4 Marks |

### Functional and Non-Functional Requirements:

Created a comprehensive list of requirements to define the functionality and performance expectations of EduGenie, an AI-powered educational platform. The requirements ensure that the system effectively supports students by providing personalized learning assistance while maintaining reliability, security, and usability.

- Functional Requirements (FR): Define the core features and services that EduGenie must provide to support students in their learning journey *(What the system should do).*
- Non-Functional Requirements (NFR): Define the quality attributes and performance standards that EduGenie must satisfy to ensure an efficient, secure, reliable, and user-friendly experience *(How the system should perform).*

Complete the descriptions and relevant details for each category to clearly specify the expected behavior and quality of the EduGenie platform.

### Step 1: Functional Requirements (FR)

| S.No | Requirement Category | Requirement Description | Priority (High/Medium/Low) |
|---|---|---|---|
| 1 | Authentication | User registration and login using JWT secure tokens and cookies. | High |
| 2 | Authorization levels | Standard user profile and administrative settings configurations. | High |
| 3 | External interfaces | Integration with Google Gemini API and local transformer models. | High |

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| 4 | Transactions processing | RAG document parsing, text chunking, and contextual query processing. | High |
| 5 | Reporting | Generating quizzes and exporting quiz results as downloadable PDF files. | Medium |
| 6 | Business rules, etc. | Engagement gamification logic: tracking daily login streaks and badges. | Medium |
| 7 | Compliance to laws | Secure environment configurations (secrets stored in .env, no plain text API keys). | High |
| 8 | Other | User query history and bookmarking response content for easy studying. | Medium |

**Step 2: Non-Functional Requirements (NFR)**

| **S.No** | **NFR Category** | **Requirement Description** | **Target Metric / Acceptance Criteria** |
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| 1 | Performance & Speed | Fast API responses, streaming AI answers under 1.5 seconds. | Page loads and streams in < 2 seconds |
| 2 | Scalability | Handles local concurrency of up to 50 active students on development server. | Supports concurrent RAG loads |
| 3 | Security & Data Privacy | Hashed passwords using bcrypt and session protection with signed JWT. | Secure local data directories |
| 4 | Reliability & Availability | Automatic model fallback from local LaMini to Gemini API if local model is offline. | 99.9% app uptime locally |
| 5 | Usability & Accessibility | Clean glassmorphism dashboard UI styled using vanilla CSS, fully responsive. | Responsive layout on all devices |
| 6 | Other | Extensible codebase modularity and reusable services (auth, RAG, AI connection). | Modular folder structure |